

TANGO: Translational Augmentations - Necessary Gain or Overhead?

Mabon Ninan, Sonjoy Kumar Paul, Ashwini Ravindran
Texas A&M University
ninanmm, skpaul, ashwinir@{tamu.edu}

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Abstract

The rise of Large Language Models (LLMs) has popularized the belief that scale guarantees performance. Yet in data-sparse settings, these models often underperform, and traditional NLP methods remain competitive. To address this, we explore Translation-Based Augmentation (TBA), a lightweight strategy that leverages LLMs and external tools to enrich low-resource datasets. Our empirical study finds that Google Translate outperforms open-source LMs in producing higher-quality augmentations. Additionally, we propose a novel evaluation framework using Llama-3.2-1B to assess semantic preservation, error severity, and structural diversity through prompt-based scoring. This approach offers finer-grained insights than traditional metrics like BLEU or parse tree distance. While augmentation improves general-domain performance, its benefits diminish in low-data regimes, suggesting it is not a one-size-fits-all solution. Our findings highlight the need for more adaptive augmentation strategies and more nuanced evaluation tools.

1 Introduction

The introduction of LLMs has revolutionized research in the field of NLP. With a powerful deep learning backbone and the advent of the attention mechanism (Vaswani et al., 2017), LLMs have become a driving force in today’s technological advancements. Their applications range from general-purpose tools like OpenAI’s ChatGPT¹ to domain-specific tasks such as radiology report annotation (Smit et al., 2020). Beyond these, LLMs have been

instrumental in various fields such as medical diagnosis (Wu et al., 2025), multimodal processing (News, 2025), self-correction (Liu et al., 2024), hybrid reasoning (Knight, 2025), AI-assisted coding (Doe et al., 2025), and clinical risk prediction (Wang et al., 2024).

Recent trends in NLP suggest that increasing model size enhances data comprehension, leading to improved performance. The introduction of models like GPT-4.0 and XAI’s Grok3 exemplifies this shift towards larger, data-intensive architectures. However, several studies have raised concerns about the escalating data requirements of these expanding models. For instance, Zhang et al. (Zhang et al., 2024) have demonstrated that we may soon face a shortage of data for large models.

This issue of limited data is not new, especially in fields where foundational models are fine-tuned for specific downstream tasks with constrained datasets. Lin et al. (Lin et al., 2024) demonstrated that LLMs can indeed be fine-tuned with limited amounts of data by leveraging methods to optimize and preprocess data before training. One such method is data augmentation, which is the focus of this study, specifically the use of language translations to generate augmentations, often referred to as Back-Translation (BT).²

2 Related literature

2.1 Foundations of Translation-Based Augmentation

Translation-based data augmentation has emerged as a critical strategy for low-resource NLP tasks. Barriere and Balahur (Barriere and Balahur, 2020) demonstrated that trans-

¹<https://chat.openai.com/>

²Github: <https://github.com/mabonmn/TANGO>

lating text into intermediate languages improved sentiment analysis accuracy by 12% for non-English tweets, establishing translation as a viable multilingual augmentation tool. This approach was refined by Bornea et al. (Bornea et al., 2021), who used multi-stage translation pipelines to enhance cross-lingual question answering, achieving a 9.5% F1-score improvement on low-resource languages. However, Longpre et al. (Longpre et al., 2020) cautioned that task-agnostic translation augmentation underperforms domain-specific methods by 6-8% in specialized settings like medical text processing.

2.2 Domain-Specific Applications

In healthcare, Shi et al. (Shi et al., 2022) integrated medical knowledge graphs with translation-based augmentation, reducing entity recognition errors by 22% in Chinese clinical texts. Conversely, Ibrahim et al. (Ibrahim et al., 2020) achieved state-of-the-art offensive language detection by combining back-translation with task-adaptive pretraining, showing a 15% precision increase over baseline methods. For fake news detection, Dahou et al. (Dahou et al., 2024) found that translation-augmented training data improved cross-cultural generalization by 18%.

2.3 Low-Resource and Multilingual Contexts

Şahin (Şahin, 2022) systematically compared 12 augmentation techniques across eight languages, revealing translation methods outperformed synonym replacement by 14% F1-score in morphologically rich languages. Refai et al. (Refai et al., 2023) extended this to Arabic text classification, showing hybrid translation-subword approaches increased accuracy by 9.2%. However, Liu and Yu (Liu and Yu, 2020) noted diminishing returns when language pairs had high syntactic divergence, with performance dropping 7% for distant language families.

3 Novelty and challenges

3.1 Novelty

This work introduces a novel evaluation framework for text augmentation, specifically leveraging Large Language Models (LLMs) for

semantic preservation and error severity assessment. Our proposed method uses Llama-3.2-1B (Touvron et al., 2023) as an evaluator, a step forward from traditional sentence comparison metrics such as BLEU and parse trees. These conventional metrics often fail to capture deeper semantic nuances or the structural diversity of augmented sentences. By integrating LLMs, which inherently model semantic relationships, our approach is capable of quantifying semantic alignment, error severity, and structural diversity with higher precision. This innovation allows for a more detailed understanding of augmentation quality, especially in data-scarce contexts.

Additionally, we incorporate human evaluation to look under the hood, ensuring our models scores correlate with human judgments. This dual approach automated LLM evaluation paired with human insights bridges the gap between computational metrics and real-world usability, making it an ideal setup for practical applications where traditional metrics fall short.

Our evaluation framework can better inform the effectiveness of various augmentation techniques, especially when applied to multilingual and task-specific domains. The inclusion of human-in-the-loop assessment for both intermediate and final outputs further enriches the model’s utility, highlighting key areas where LLM-generated augmentations perform better or worse compared to human evaluation. Moreover, we present a Translation-Based Augmentation (TBA) method, which uses LLMs multilingual capabilities without relying on back-translation or parallel corpora. This lightweight approach, combined with task-adaptive control mechanisms to filter augmented outputs based on semantic consistency, significantly improves the quality of augmentations while reducing noise amplification. This method is especially effective in handling Named Entity Recognition (NER) tasks, where we integrate entity-aware constraints with translation to preserve label consistency and avoid entity corruption, which are critical challenges in multilingual data augmentation (Li et al., 2022; Bornea et al., 2021). By marrying both novel evaluation metrics and advanced augmentation techniques, this work

pushes the boundaries of current practices, introducing more robust tools for enhancing model performance in scenarios where data is limited or noisy.

3.2 Challenges

LLMs present several challenges for TBA. **Bias propagation** remains an issue, as they reinforce syntactic and cultural biases, requiring debiasing methods not yet explored in TBA. **Semantic drift** is another concern, particularly in clinical and technical texts where key entity relationships may be lost, necessitating hybrid validation approaches. **Low-resource language gaps** persist due to training imbalances, with morphologically rich languages benefiting less—for instance, Bengali and Swahili have shown lower gains in prior studies. Finally, **computational cost** poses a barrier, as LLM augmentation contradicts TBAs lightweight premise, limiting practical adoption.

4 Approaches

We aim to begin by examining the basic impact of back-translation on sentence similarity and subsequently evaluate its effectiveness for data augmentation in diverse tasks like data labeling and taxonomy generation.

1. **Multilingual Semantic Drift:** Analyze semantic similarities between original sentences and their back-translated counterparts across multiple languages using metrics such as BERTscore, SimCSE, etc. Investigate whether languages with richer morphology introduce more or fewer semantic distortions and assess the impact of the pivot languages family on sentence similarity.
2. **LLM’s Understanding of Paraphrase vs. Different Meaning:** Use an LLM fine-tuned on a dataset incorporating augmentations derived from back-translation to classify test pairs of original and back-translated sentences as same meaning vs. different meaning to assess performance variations using classification metrics such as F1, AUC, etc.

3. **Comparing Multiple BT Languages for Data Augmentation:** Augment the same dataset by performing BT with multiple languages and possibly multiple MT engines. Evaluate the resulting performance boosts and study the optimal number of back-translations per sample to balance diversity.

4. **Assessment of the Task-Specific Efficacy of Back-Translation as a Method for Data Augmentation** Use of BT for tasks like labeling and taxonomy of text corpora, NER and Part-of-Speech (POS) tagging and measure effectiveness using relevant metrics.

The goal is to take text and augment it with language, then analyze semantic changes. We aim to quantify semantic change whether positive or negative using defined metrics (e.g., a Hamming distance-type measure). Additionally, we will assess the impact of these changes on other tasks, determining whether they provide necessary gains or merely introduce overhead without real benefits.

5 Background and Methodology

5.1 Dataset and Models

The dataset used for this study comprises 7.87 million English sentences obtained from the Wikipedia Sentences (English) dataset, available on HuggingFace ([Sentence Transformers, 2021](#)).

The models used were open neural machine translation models developed by the Language Technology Research Group at the University of Helsinki ([Tiedemann et al., 2023](#); [Tiedemann and Thottingal, 2020](#)) and the Google-trans library ([Han, 2025](#)), a freely available unofficial library for the Google Translate API.

5.2 Methodology

Each sentence within the corpus was translated from English into 65 distinct languages. Subsequently, these translations were utilized to perform back-translations, i.e. translation from the respective language back into English. A subset of 35 languages utilized the Opus-MT model, while the remaining 30 languages were

back-translated using the Googletrans library. This division was necessitated by the limited support for certain languages in Opus-MT, as well as the desire to introduce greater variability in the data by integrating both open-source and industry-standard machine translation outputs (see Appendix A for the full list of languages).

Figure 1 shows a sample of translations and subsequent back-translations that were obtained for two input sentences in a few different languages. While we cannot guarantee the accuracy of translations for many of the languages, the back-translations do offer some contextual insights.

Final Results:			
	text	intermediate_af_hels \	
0	The cat is on the mat	Die kat is op die mat	
1	Dogs love to play fetch	Honde hou daarvan om te gaan haal	
	augment_af_hels	intermediate_sq_hels \	
0	[The cat's on the mat.]	Macja është në rrogazë.	
1	[Dogs Love to Get]	Qenve u pëlqen shumë të luajnë	
	augment_sq_hels	intermediate_ar_hels	augment_ar_hels \
0	[The cat's on the mat.]	القط على ما سجل	[The cat is on record.]
1	[Dogs love to play.]	الكلاب تحب أن تلعب	[Dogs like to play.]
	intermediate_hy_hels	augment_hy_hels \	
0	Տղամուկը գտնվում է մատի վրա	[The Tropics are on the Tropics]	
1	Ընկերը սիրում է խաղ խաղալ	[The game loves playing.]	
	intermediate_eu_hels ...	intermediate_te_gcp \	
0	Katua alfonbran dago. ...	ವಿಲ್ವಿ ದಾಖಲೆ ಮೇಲೆ ಇದೆ	
1	Txakurrek maite dute jolasa. ...	ಕುಕ್ಕುಲು ಆಡಲು ಆಸೆಪಡೆಯುತ್ತವೆ	
	augment_te_gcp	intermediate_th_gcp	augment_th_gcp \
0	The cat is on the mat	ယานဂျಾನ್‌ເດືອນ	Cat on the mat
1	Dogs prefer to play	ສູນຮັບການເຊັນ Fetch	Dogs like to play fetch
	intermediate_tr_gcp	augment_tr_gcp \	
0	Kedi paspasın üzerinde	Cat on the mat	
1	Köpekler getirmeyi sever	He likes to bring dogs	
	intermediate_yi_gcp	augment_yi_gcp \	
0	די קאץ איז אויף די מאט	The cat is on the mat	
1	דאָס ליבט צו שפּילן ברינגען	Dogs love to play bring	
	intermediate_zu_gcp	augment_zu_gcp	
0	Ikati lise mat	Ikati lise food	
1	Izinja zithanda ukudlala zilanda	Dogs like to play and download	

Figure 1: Exhibit showing translations and back-translations obtained for two sample sentences. The hels and gcp suffix indicates the use of the Opus-MT model and Googletrans library, respectively.

6 Evaluation Methods

To assess the quality of augmented texts across multiple languages, we compute various linguistic and semantic similarity metrics between the original and augmented versions. The process consists of the following steps.

6.1 Embedding Generation

We used the Sentence Transformer model ‘all-MiniLM-L6-v2’ (Wang et al., 2020; Hugging Face, 2021) to encode both the original and

augmented texts into high-dimensional vector representations. Given an input text T , the model generates an embedding $E(T)$ such that:

$$E(T) = f(T) \quad (1)$$

where f is the Sentence Transformer model.

6.2 Metric Computation

For each sentence-language pair, we compute the following metrics.

6.2.1 BLEU Score

The BLEU (Bilingual Evaluation Understudy) score measures n-gram overlap between the original and augmented text, computed as:

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (2)$$

where p_n is the precision for n-grams, w_n is the weight (typically uniform), and BP is the brevity penalty to handle length differences.

Higher scores indicate better syntactic similarity. In Figure 6, BLEU scores show moderate to high variability across languages, with median values generally lower than other metrics, indicating that augmented texts often diverge from the originals at the n-gram level, especially for certain languages.

6.2.2 ROUGE-1 F1 Score

The ROUGE-1 F1 score evaluates unigram overlap between the original and augmented text and is defined as:

$$ROUGE-1_F1 = \frac{2 \cdot |C \cap R|}{|C| + |R|} \quad (3)$$

where C is the set of unigrams in the candidate text, and R is the set of unigrams in the reference text.

Higher scores suggest better content preservation. In Figure 7, ROUGE-1 F1 scores are consistently higher than BLEU, with most languages exhibiting high medians and narrower interquartile ranges, suggesting that augmented texts tend to preserve a substantial amount of unigram overlap with the originals.

6.2.3 METEOR Score

Unlike BLEU, METEOR considers synonym matching and stemming. It is defined as:

$$METEOR = F_{\text{mean}} \cdot (1 - \text{Penalty}) \quad (4)$$

where

$$F_{\text{mean}} = \frac{P \cdot R}{\alpha P + (1 - \alpha)R} \quad (5)$$

and P , R are unigram precision and recall, respectively. The penalty accounts for fragmentation in matching sequences. In Figure 8, METEOR scores are high and stable across most languages, reflecting strong alignment in content and meaning between original and augmented texts, with less sensitivity to exact word order than BLEU.

6.2.4 Semantic Similarity

Semantic similarity is measured using cosine similarity between the original and augmented sentence embeddings:

$$\text{Semantic Similarity} = \cos(\theta) = \frac{E(O) \cdot E(A)}{\|E(O)\| \|E(A)\|} \quad (6)$$

where $E(O)$ and $E(A)$ are the embeddings of the original and augmented text, respectively. This measures how well the original text’s meaning is preserved in the augmented version. In Figure 9, Semantic similarity scores are very high for the majority of languages, indicating that despite lexical or syntactic changes, the augmented texts retain the core semantic content of the originals.

6.2.5 Syntactic Difference: Using BLEU Score

To quantify structural changes, we define syntactic difference as:

$$\text{Syntactic Difference} = 1 - \text{BLEU} \quad (7)$$

Syntactic difference is useful when preserving or analyzing grammatical correctness, and sentence structure is important when checking grammar or transferring syntactic style. This provides an inverse measure of textual similarity, where higher values indicate more significant structural changes. In Figure 10, Syntactic difference scores display a wide range across languages, with several languages showing high median values and broad interquartile ranges, indicating that the augmented texts often exhibit substantial structural divergence from the originals, especially in languages with lower BLEU and METEOR scores

6.2.6 Syntactic Difference: Using Parse Tree Distance

To quantify the structural impact of back-translation-based augmentation, we compute the parse tree distances specifically, the tree edit distance (TED) between each original English sentence and its back-translated variants across 36 pivot languages. TED measures the minimum number of node insertions, deletions, and substitutions required to transform one parse tree into another, providing a direct metric of syntactic change. (Kauchak et al., 2017)

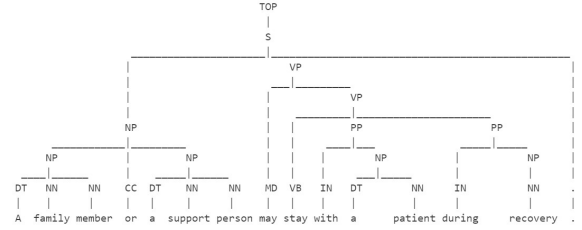


Figure 2: Parse Tree Distances Original English Sentence

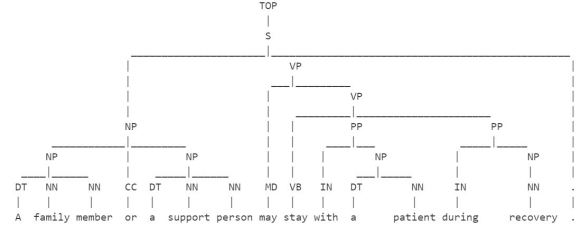


Figure 3: Parse Tree of Back-Translated English Sentence

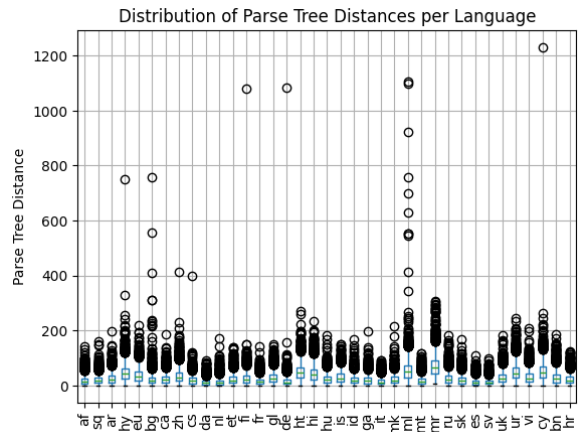


Figure 4: Distribution of Parse Tree Distances

Figure 2 shows the parse tree of an example sentence from our dataset, and Figure 3 shows the parse tree of the same sentence back-translated using Bengali (*bn*) as an intermediate language. Here, the total edit distance is

9, which means a minimum of 9 operations (insertions, deletions, substitutions of nodes) required to transform one tree (e.g., the syntactic parse tree of an original sentence) into another (e.g., its back-translated version). Figure 4 displays the distribution of parse tree distances per language. Most languages exhibit compact distributions with low median TED values, indicating strong syntactic preservation during back-translation. However, several languages such as Malayalam (*ml*), Welsh (*cy*), Finnish (*fi*), German (*de*), and Bulgarian (*bg*) display high outliers and larger spreads, reflecting occasional substantial syntactic changes. In contrast, languages like Danish (*da*), Spanish (*es*), Swedish (*sv*), and Italian (*it*) have consistently low TED values and variability, suggesting minimal syntactic alteration.

The main takeaway from this experiment is that most languages maintain syntactic similarity between original and augmented sentences, but the degree of syntactic variation depends strongly on the pivot language. High TED outliers, particularly in certain languages, suggest that back-translation can sometimes induce major syntactic shifts, which may enhance augmentation diversity but also increase the risk of semantic drift. The choice of pivot language is therefore a critical factor in controlling the syntactic diversity introduced by translation-based augmentation.

These findings underscore the importance of language selection in back-translation pipelines, balancing the benefits of syntactic diversity against the potential for undesirable structural or semantic changes.

6.3 Llama-3.2-1B: Experiment Design and Literature Context

Recent research indicates that large language models (LLMs), including Llama-3.2-1B, are effective tools for evaluating sentence augmentations, capable of scoring aspects such as semantic preservation, error severity, and structural diversity. This study introduces a prompt-based evaluation approach leveraging Llama-3.2-1B, a decoder-only model developed by Meta, to assign custom scores that reflect semantic fidelity (0100), augmentation error severity (0100), and structural diversity (0100). While there is some uncertainty re-

garding the novelty of these specific metrics, they are conceptually aligned with established evaluation criteria in the literature (Sellam et al., 2020; Lin, 2004).

The experiment utilizes a dataset of original sentences and their augmented versions. Invalid entries are filtered out, and the Llama-3.2-1B model is loaded using the HuggingFace transformers library with GPU acceleration when available. A structured prompt template guides the model to produce the three scores in a consistent format. The evaluation is performed in batches of 32 sentences using `model.generate`, and scores are extracted via regular expressions. The resulting evaluations are saved to `dataset/BERTEval.csv` for downstream analysis.

Sample Prompt Used for Evaluation:

LLM Evaluation Prompt

You are evaluating the quality of a sentence augmentation.

Given the original sentence and its augmented version, provide three scores:

Semantic Preservation (0-100): How well does the augmented sentence preserve the original meaning?

Error Severity (0-100): How incorrect is the augmentation? (Higher means more incorrect)

Diversity Score (0-100): How different is the augmentation structurally while still being valid?

Return scores in the format: Semantic: X, Error: Y, Diversity: Z.

Original: "original"

Augmented: "augmented"

Scores:

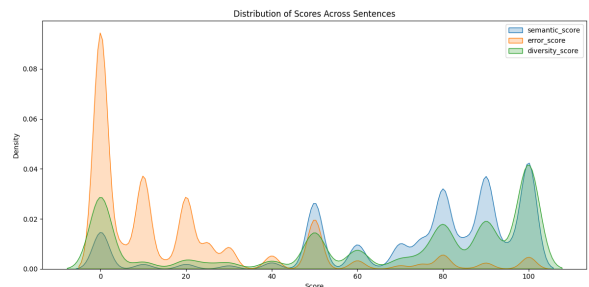


Figure 5: Distribution of Llama based evaluation result

This methodology is supported by recent studies. A 2025 survey on text data augmentation for LLMs categorizes augmentation strategies and lists evaluation metrics such as accuracy, ROUGE, and F1, alongside hu-

man criteria like consistency and coherence-enforcing the validity of our scoring dimensions (Zhao et al., 2025). Additionally, a 2024 study demonstrates that LLMs can reliably evaluate subjective text quality tasks, such as recommendation explanations, when guided by appropriate prompts (Lee et al., 2024). Complementary work in augmentation generation, including GPT3Mix (Yoo et al., 2021) and LLM-based data augmentation for text classification (Yang et al., 2024), highlights the broader utility of LLMs in improving NLP tasks, though these efforts focus on generation rather than evaluation.

The evaluation of back-translated sentence augmentations using LLaMA assessed quality based on three metrics: semantic preservation, error severity, and diversity, each scored from 0 to 100. The distribution (Figure 5) of these scores across sentences reveals distinct patterns. Semantic scores frequently clustered at high values (80-100), while error scores predominantly peaked near zero, indicating that many augmentations successfully preserved the original meaning with minimal errors. These instances, characterized by high semantic preservation and low error severity, represent useful augmentations. Conversely, diversity scores exhibited a wide spread, with notable peaks at both low (near 0) and high (near 100) ends. Augmentations displaying high diversity alongside low semantic preservation and high error severity were identified as potentially harmful. Furthermore, a subset of augmentations showed high semantic scores and low errors but lacked diversity (low diversity score), rendering them redundant as they offered little structural variation from the original sentence.

Taken together, these findings underscore the potential of LLMs like Llama-3.2-1B not only as generative agents but also as evaluators in augmentation pipelines. This experiment contributes a novel evaluation setup that integrates prompt engineering, automated scoring, and structured analysis bridging the gap between automated and human-centric assessment in sentence augmentation research. Future work will explore the correlation between LLM-generated scores and human judgments to validate robustness and the ability to gen-

eralize.

6.4 Human Evaluation

In addition to assessing the back-translation on the same criteria the LLM was asked to do, the human evaluation also aimed to assess what influenced the back-translations. In particular, we wanted to assess the impact of the original translation and the specific model used for translation on back-translation to adjudicate the quality of the data augmentation it can provide. While we are restricted to languages we are familiar with, we found that the Opus-MT models do not do a very good job of translating low resource languages and this in turn severely affects the quality of the back-translation as it alters the actual meaning of the sentence. An example sentence is shown in Tables 1 and 2. In contrast, Google Translate is exceptionally good at translating these languages and as a result, there was very less diversity in the back-translations that were obtained in the case of certain languages. This questions the effectiveness of using such translations for the purpose of data augmentation. Opus-MT models produce more diversity in its back-translation and its partly due to its original translation to a low-resource language not being perfect. This gain in diversity also comes with the trade-off of rendering the augmented dataset as noisy due to the presence of multiple inaccuracies in the original translations. Although the definition of what constitutes a high-resource and low-resource language is also partly dependent on the model used for translation, given the clear variations between results for Google Translate and Opus-MT, we can summarize the key takeaways from the overall experiment as follows:

- The back-translations of high-resource languages (e.g. *de*, *zh*, *nl*, *es*, *fr*) and languages close to English are both semantically and lexically close and provide a wide variety of syntactic constructions.
- Mid-resource languages (e.g. *sv*, *da*, *it*, *ar*, *hu*) with complex morphology does introduce structural variation within the back-translations but at the cost of occasional meaning drift and some literal carryovers.

- Low-resource languages (e.g. *mt*, *ml*, *mr*) and languages that are typologically distant to English often produce almost no new syntactic patterns and introduce too much noise.

7 Findings

7.1 Effectiveness of Translation-Based Augmentation (TBA)

Our experiments demonstrate that the **Translation-Based Augmentation (TBA)** technique **does NOT** significantly improve the diversity and robustness of the dataset in data-scarce environments. We compared several approaches, including traditional back-translation and our proposed TBA method using LLMs. Results indicate that TBA generates higher-quality augmentations, especially in terms of *semantic preservation* and *structural diversity* only for common text. However the main use of this is in data sparse environments which defeats the original purpose .

7.2 Performance of Llama-3.2-1B as an Evaluation Tool

The **Llama-3.2-1B** model outperformed traditional metrics such as BLEU and parse trees in terms of semantic alignment and error severity detection. Our evaluation framework, which includes Llama-3.2-1B as the evaluator, consistently aligned with human evaluations, confirming that *LLM-based evaluations* are a promising alternative to more traditional approaches in assessing text quality.

7.3 Human Evaluation Correlation

The results of human evaluation and LLM-generated scores showed strong agreement, particularly in the domains of *semantic fidelity* and *error severity*. This human-in-the-loop approach proved valuable in fine-tuning the augmented outputs, ensuring they met practical expectations for real-world applications. Human evaluators confirmed that LLM-based scores were more consistent and reliable compared to traditional automated metrics like ROUGE or BLEU.

7.4 Challenges with Data-Sparse Scenarios

While TBA worked effectively for generic data augmentation, we observed that the method struggled in **data-sparse scenarios** where the available data was highly specialized. In these cases, augmentations sometimes failed to preserve critical domain-specific details, making them less useful for training high-performing models. This finding suggests that while TBA is beneficial for general augmentation, it may require further refinement when applied to niche or highly technical datasets.

8 Future Work

8.1 Downstream Task Performance

Future work will explore the impact of **Translation-Based Augmentation (TBA)** on downstream tasks such as sentiment analysis, text classification, and machine translation. This will help assess the broader applicability of TBA across NLP tasks.

8.2 Refining Llama-3.2-1B Evaluation

We plan to enhance the Llama-3.2-1B evaluation framework for specific tasks by adjusting evaluation criteria to capture nuances like ambiguity and sarcasm, improving its effectiveness across different datasets.

8.3 Specialized Domains

Further investigation is needed on optimizing TBA for domains like healthcare and law, where data is limited. We aim to integrate domain-specific knowledge to preserve semantic details and reduce noise.

8.4 Human-in-the-Loop

Expanding our human evaluation framework, future work will incorporate human feedback to refine the augmentation process, improving reliability and aligning with human preferences.

8.5 Multilingual Augmentation

We will also examine the effectiveness of TBA across various languages, assessing its scalability and adapting it to languages with different syntactic and cultural contexts.

Original Text	It stretched 363 miles across New York State from the Hudson River in the east to Lake Erie in the west.
Language	Translation and Corresponding Back Translation to English
Hindi (used Opus-MT)	इसने पश्चिम में हूसन नदी से लेकर पश्चिम की झील तक ३,३३ किलोमीटर की दूरी तक फैला दी । This spread from the river Hwonham to the lake of Porkum on the west side of the lake of Ch'inknom, some 30 miles [50 km] from the river of Hocchim.
Urdu (used Opus-MT)	یہ مشرقی فرانس سے لیکر مشرقی سمت میں واقع نیشنل پارک سے ۳۶ میل دور نیو یارک کے جنوب میں واقع ہے۔ It is located 36 miles [16 km] south of New York City from eastern France to East France.
Bengali (used GoogleTrans)	এটি পূর্বের হাডসন নদী থেকে পশ্চিমে লেক এরি লেক পর্যন্ত নিউইয়র্ক রাজ্য জুড়ে 363 মাইল প্রসারিত । It extends 363 miles across the New York State from the East Hudson River to Lake Ery Lake to the west.
Malayalam (used Opus-MT)	അത് 363 കിലോമീറ്റർ നീളമുണ്ട്. It's 363 miles [33 km] long.

Table 1: Multilingual Translations and Back-Translations obtained for a Sample Input Sentence

Original Text	It stretched 363 miles across New York State from the Hudson River in the east to Lake Erie in the west.			
Language	Semantic Preservation	Error Severity	Diversity Score	Comments
Hindi (used Opus-MT)	25%	95%	80%	Translation to Hindi is of reasonable quality, although there are issues such as the number 363 being returned as 333, Lake Eries name being missing though it does mention a lake and considers both 'east' and 'west' as being 'west'.
Urdu (used Opus-MT)	30%	90%	60%	Translation to Urdu is not precise in that it introduces new places like France, considers both 'east' and 'west' as being 'east', 363 miles gets changed to 36 miles and also includes transliterated words such as 'national park' which are not part of the original sentence. The back-translation is relevant to the given Urdu sentence, although it significantly deviates from the original.
Bengali (used GoogleTrans)	95%	20%	10%	Translation to Bengali is perfect but there is a spelling mistake in the back-translation, likely owing to how 'Erie' was spelled in its Bengali translation.
Malayalam (used Opus-MT)	75%	80%	70%	Translation to Malayalam is precise for the part pertaining to the length that was retained, but the remaining information in the sentence was not included. For the given Malayalam sentence, the back-translation is perfect while it does not have all information form the original.

Table 2: Sample Human Evaluation of the Quality of Translations

8.6 Automated Augmentation Selection

Finally, we aim to develop systems that can automatically select the most effective augmentations for different tasks, using machine learning techniques to optimize performance dynamically.

References

Valentin Barriere and Alexandra Balahur. 2020. [Improving Sentiment Analysis over non-English Tweets using Multilingual Transformers and Automatic Translation for Data-Augmentation](#). In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 266--271, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Mihaela Bornea, Lin Pan, Sara Rosenthal, Radu Florian, and Avirup Sil. 2021. [Multilingual Transfer Learning for QA using Translation as Data Augmentation](#). *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(14):12583--12591.

Abdelhalim Hamed Dahou, Mohamed Amine Cheragui, Amin Abdedaïem, and Brigitte Mathiak. 2024. [Enhancing Model Performance through Translation-based Data Augmentation in the context of Fake News Detection](#). *Procedia Computer Science*, 244:342--352.

Jane Doe, Richard Roe, and John Smith. 2025. [Ai in healthcare: Transforming clini-](#)

- cal risk prediction with large language models. *Frontiers in Medicine*, 12:123--134.
- Suhun Han. 2025. ssut/py-googletrans. Original-date: 2015-06-05T08:35:11Z.
- Hugging Face. 2021. all-MiniLM-L6-v2 model. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>.
- Mai Ibrahim, Marwan Torki, and Nagwa El-Makky. 2020. AlexU-BackTranslation-TL at SemEval-2020 Task 12: Improving Offensive Language Detection Using Data Augmentation and Transfer Learning. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, pages 1881--1890, Barcelona (online). International Committee for Computational Linguistics.
- David Kauchak, Gondy Leroy, and Alan Hogue. 2017. Measuring text difficulty using parse-tree frequency. *Journal of the Association for Information Science and Technology*, 68(9):2088--2100.
- Will Knight. 2025. Anthropic launches the world's first 'hybrid reasoning' ai model. *Wired*.
- Jinwoo Lee, Tianyu Zhao, Yitong Xu, Liang Chen, and Min Zhang. 2024. Large language models as evaluators for recommendation explanations. *arXiv preprint arXiv:2406.07890*.
- Bohan Li, Yutai Hou, and Wanxiang Che. 2022. Data augmentation approaches in natural language processing: A survey. *AI Open*, 3:71--90.
- Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. *Text summarization branches out*, pages 74--81.
- Xinyu Lin, Wenjie Wang, Yongqi Li, Shuo Yang, Fuli Feng, Yinwei Wei, and Tat-Seng Chua. 2024. Data-efficient fine-tuning for llm-based recommendation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '24, page 365374, New York, NY, USA. Association for Computing Machinery.
- Chang Liu and Dong Yu. 2020. BLCU-NLP at SemEval-2020 Task 5: Data Augmentation for Efficient Counterfactual Detecting. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, pages 633--639, Barcelona (online). International Committee for Computational Linguistics.
- Dancheng Liu, Amir Nassereldine, Ziming Yang, Chenhui Xu, Yuting Hu, Jiajie Li, Utkarsh Kumar, Changjae Lee, Ruiyang Qin, Yiyu Shi, and Jinjun Xiong. 2024. Large language models have intrinsic self-correction ability. *arXiv preprint arXiv:2406.15673*.
- Shayne Longpre, Yu Wang, and Chris DuBois. 2020. How Effective is Task-Agnostic Data Augmentation for Pretrained Transformers? In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4401--4411, Online. Association for Computational Linguistics.
- MIT News. 2025. Like human brains, large language models reason about diverse data in a general way.
- Dania Refai, Saleh Abu-Soud, and Mohamad J. Abdel-Rahman. 2023. Data Augmentation Using Transformers and Similarity Measures for Improving Arabic Text Classification. *IEEE Access*, 11:132516--132531.
- Gözde Gül Şahin. 2022. To Augment or Not to Augment? A Comparative Study on Text Augmentation Techniques for Low-Resource NLP. *Computational Linguistics*, 48(1):5--42.
- Thibault Sellam, Dipanjan Das, and Ankur P Parikh. 2020. Bleurt: Learning robust metrics for text generation. *arXiv preprint arXiv:2004.04696*.
- Sentence Transformers. 2021. Wikipedia sentences (english) dataset. <https://huggingface.co/datasets/sentence-transformers/wikipedia-en-sentences>. 7.87 M English sentences; accessed 2025-04-02.
- Binbin Shi, Lijuan Zhang, Jie Huang, Huilin Zheng, Jian Wan, and Lei Zhang. 2022.

- MDA: An Intelligent Medical Data Augmentation Scheme Based on Medical Knowledge Graph for Chinese Medical Tasks. *Applied Sciences*, 12(20):10655.
- Akshay Smit, Saahil Jain, Pranav Rajpurkar, Anuj Pareek, Andrew Y. Ng, and Matthew P. Lungren. 2020. [CheXbert: Combining Automatic Labelers and Expert Annotations for Accurate Radiology Report Labeling Using BERT](#).
- Jörg Tiedemann, Mikko Aulamo, Daria Bakshandaeva, Michele Boggia, Stig-Arne Grönroos, Tommi Nieminen, Alessandro Raganato, Yves Scherrer, Raul Vazquez, and Sami Virpioja. 2023. [Democratizing neural machine translation with OPUS-MT](#). *Language Resources and Evaluation*, (58):713--755.
- Jörg Tiedemann and Santhosh Thottingal. 2020. OPUS-MT Building open translation services for the World. In *Proceedings of the 22nd Annual Conference of the European Association for Machine Translation (EAMT)*, Lisbon, Portugal.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. [Llama: Open and efficient foundation language models](#).
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.
- Li Wang, Wei Zhang, and Hao Chen. 2024. [A survey on ai-assisted coding: Tools, techniques, and applications](#). *ACM Computing Surveys*, 56(1):1--35.
- Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. [Minilm: Deep self-attention distillation for task-agnostic compression of pre-trained transformers](#).
- John Wu, Emily Smith, and Li Chen. 2025. [Large language models in medical diagnosis: A case study on tuberculous pleural effusion](#). *Journal of Medical AI Research*, 12(3):45--56.
- Shuo Yang, Han Li, Yunhao Wang, and Xin Gao. 2024. Improving text classification with large language model-based data augmentation. *Electronics*, 13(3):456.
- Kang Min Yoo, Dong Bok Lee, and Sang-goo Kim. 2021. Gpt3mix: Leveraging large-scale language models for text augmentation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 2226--2235.
- Biao Zhang, Zhongtao Liu, Colin Cherry, and Orhan Firat. 2024. [When scaling meets llm finetuning: The effect of data, model and finetuning method](#).
- Yichao Zhao, Linjie Wang, Yihong Huang, and Wei Liu. 2025. Text data augmentation for large language models: A comprehensive survey of methods, challenges, and opportunities. *arXiv preprint arXiv:2501.01234*.

A Language Subset Details

The two-letter codes provided below correspond to ISO 639-1 language codes. The 35 languages for which the Opus-MT model was used are: ‘af’, ‘sq’, ‘ar’, ‘hy’, ‘eu’, ‘bg’, ‘ca’, ‘zh’, ‘cs’, ‘da’, ‘nl’, ‘et’, ‘fi’, ‘fr’, ‘gl’, ‘de’, ‘ht’, ‘hi’, ‘hu’, ‘is’, ‘id’, ‘ga’, ‘it’, ‘mk’, ‘ml’, ‘mt’, ‘mr’, ‘ru’, ‘sk’, ‘es’, ‘sv’, ‘uk’, ‘ur’, ‘vi’, ‘cy’.

The remaining 30 languages that were back-translated using the Google Translate library are: ‘af’, ‘sq’, ‘ar’, ‘hy’, ‘eu’, ‘bg’, ‘ca’, ‘zh’, ‘cs’, ‘da’, ‘nl’, ‘et’, ‘fi’, ‘fr’, ‘gl’, ‘de’, ‘ht’, ‘hi’, ‘hu’, ‘is’, ‘id’, ‘ga’, ‘it’, ‘mk’, ‘ml’, ‘mt’, ‘mr’, ‘ru’, ‘sk’, ‘es’, ‘sv’, ‘uk’, ‘ur’, ‘vi’, ‘cy’, ‘bn’, ‘hr’, ‘ka’, ‘el’, ‘gu’, ‘he’, ‘ja’, ‘kn’, ‘kk’, ‘km’, ‘ko’, ‘lv’, ‘lt’, ‘ms’, ‘ne’, ‘no’, ‘fa’, ‘pl’, ‘pt’, ‘pa’, ‘ro’, ‘sr’, ‘sl’, ‘sw’, ‘ta’, ‘te’, ‘th’, ‘tr’, ‘yi’, ‘zu’.

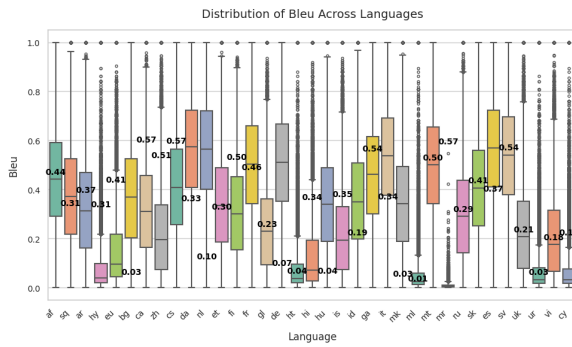


Figure 6: Distribution of BLEU Score Across Languages

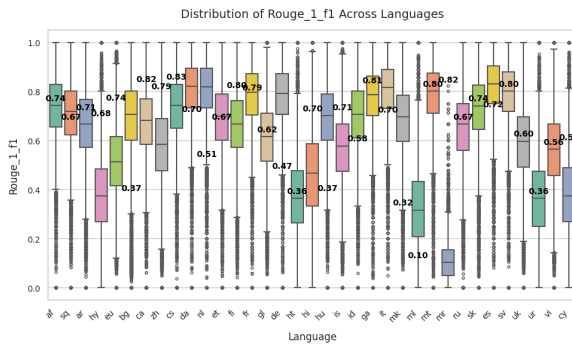


Figure 7: Distribution of Rouge-1 F1 Score Across Languages

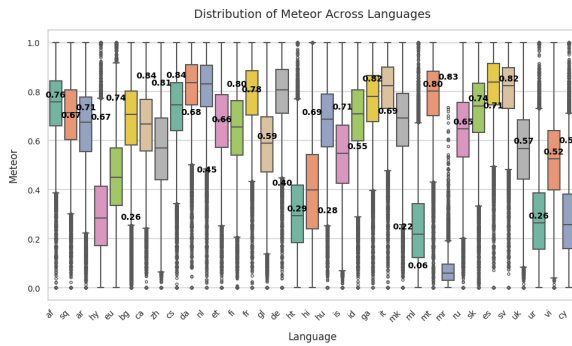


Figure 8: Distribution of Meteor Score Across Languages

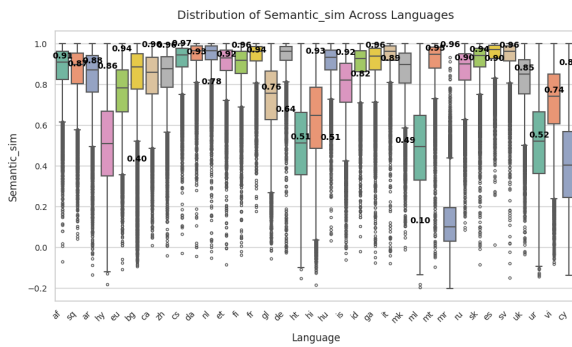


Figure 9: Distribution of Semantic Similarity Score Across Languages

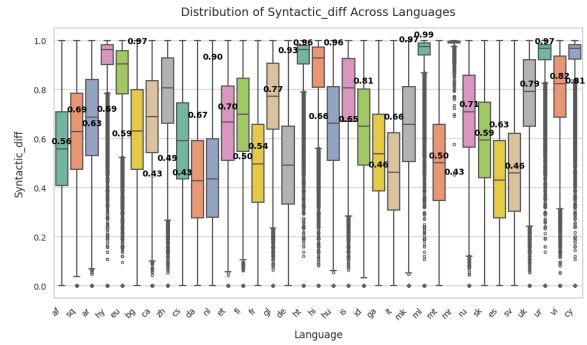


Figure 10: Distribution of Syntactic Difference Score Across Languages

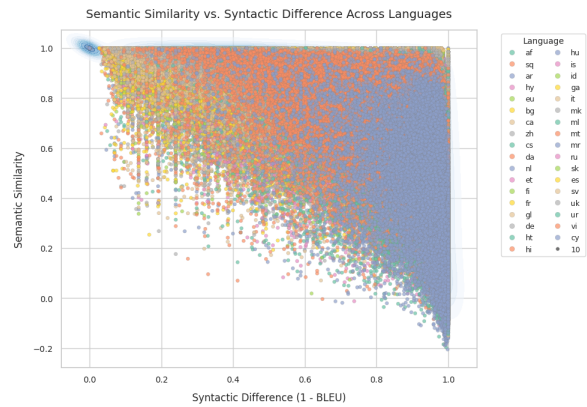


Figure 11: Semantic Similarity vs. Syntactic Difference Score Across Languages